Aiqon AI Take-Home Assignment: Technical Report

Forecasting Port Congestion and Shipping Delays Using Interpretable Machine Learning

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1. Problem Statement & Operational Objective

Global container supply chains rely on smooth maritime terminal throughput. When vessel arrival surges or adverse weather disrupt scheduled operations, queuing backlogs escalate rapidly, producing substantial ship idling costs, missed logistics handoffs, and steep carrier demurrage penalties. The objective of this project is to build an **interpretable early-warning forecasting model** that predicts whether a commercial port will experience an operational delay exceeding **12 hours** during the **next six-hour interval**, using only conditions observable at the close of the **current six-hour window**. Crucially, the system must deliver actionable probability scores, support flexible decision thresholds, and provide clear operational explanations without claiming ungrounded causal relationships.

2. Synthetic Dataset & Simulation Dynamics

To simulate realistic operational patterns without compromising proprietary port data, a synthetic logging system (data/generate_dataset.py) models 5 distinct marine terminals (PORT_A to PORT_E) over 500 consecutive six-hour intervals from January 1, 2026 to May 5, 2026 (2,500 total records). Key generative mechanisms include:

- **Port Capacities & Arrivals:** Baseline handling capacities range from 9 to 16 vessels/window with Poisson-distributed arrivals modulated by weekend factors (0.90x) and diurnal peak surges (1.12x).
- **Weather Impacts:** Stochastic weather transitions (Clear 66%, Rain 27%, Storm 7%) scale effective service capacity downward (Rain: 0.88x, Storm: 0.68x) and create arrival clustering.
- **Queue Physics:** Unserviced vessels carry over into subsequent intervals via recursive backlog tracking: $Backlog_t = max(0, Queue_t + Arrivals_t - EffectiveCapacity_t)$.
- **Delay Equation & Noise:** Service delays compound arrival pressure ratio (>0.70 load), accumulated queue length (1.15x), storm disruption shocks, and Gaussian process noise, yielding a 11.9% overall delay rate (>12h).

3. Target Definition, Alignment & Temporal Leakage Controls

Forecasting requires strict separation between the observation window and the predicted outcome horizon:

- **Target Formulation:** The target target_delayed_next_6h is defined as (actual_delay_hours > 12.0) occurring during the immediate next 6-hour operational interval ($T_{cutoff} \rightarrow T_{cutoff} + 6h$).
- **Intra-Port Alignment:** In src/prepare_data.py, target shifting is executed strictly within port groups via df.groupby('port_id')['is_delayed'].shift(-1). Cross-port boundary contamination is impossible.
- **Boundary Row Handling:** The terminal record for each port lacks a subsequent observation; exactly 5 boundary records are pruned, yielding a validated modeling dataset of 2,495 rows.
- **Zero Target Leakage:** Current-interval outcome fields (delay_hours, actual_delay_hours, is_delayed) are permanently dropped from the feature space, guaranteeing that only pre-cutoff information is visible to classifiers.

Operational Disclosure: All findings in this report derive from synthetic simulation logs. While calibrated to simulate empirical queue physics, results should not be interpreted as empirical real-world validation on commercial ports.

4. Machine Learning Pipeline & Chronological Validation

Random k-fold cross-validation is invalid for temporal logistics because randomly shuffling rows allows future patterns to leak into past predictions. We enforce a **strict chronological partition** with shared timestamp cutoffs across all 5 ports:

- **Training Split (70% — 1,745 rows):** Jan 1, 2026 00:00 to Mar 29, 2026 00:00 (204 delays, 11.7% base rate).
- **Validation Split (10% — 250 rows):** Mar 29, 2026 06:00 to Apr 10, 2026 12:00 (39 delays, 15.6% base rate) — reserved strictly for threshold tuning.
- **Held-Out Test Split (20% — 500 rows):** Apr 10, 2026 18:00 to May 5, 2026 12:00 (53 delays, 10.6% base rate) — locked for final unbiased evaluation.

Model Architecture: We deploy a scikit-learn Pipeline encapsulating a ColumnTransformer (one-hot encoding port_id, weather_condition, and day_of_week, with numeric passthrough) and a RandomForestClassifier (100 estimators, max_depth=6, min_samples_leaf=2, class_weight='balanced'). The pipeline is fitted exclusively on the training partition.

5. Empirical Performance & Baseline Benchmark

The model is compared against a zero-intelligence majority baseline (DummyClassifier, always predicting on-time). Evaluation metrics reflect the actual reproducible pipeline outputs:

Model & Policy Configuration	Split	Thresh	Accuracy	Precision	Recall	F1-Score	Delays Caught
Dummy Baseline (Always On-time)	Validation	—	84.40%	0.00%	0.00%	0.0000	0 / 39 (0.0%)
Dummy Baseline (Always On-time)	Held-out Test	—	89.40%	0.00%	0.00%	0.0000	0 / 53 (0.0%)
Random Forest (Default Baseline)	Validation	0.50	83.60%	48.00%	61.54%	0.5393	24 / 39 (61.5%)
Random Forest (Default Baseline)	Held-out Test	0.50	85.80%	35.94%	43.40%	0.3932	23 / 53 (43.4%)
Random Forest (Val F1-Maximizing)	Validation	0.55	87.20%	60.00%	53.85%	0.5676	21 / 39 (53.8%)
Random Forest (Val F1-Maximizing)	Held-out Test	0.55	87.60%	37.84%	26.42%	0.3111	14 / 53 (26.4%)
Random Forest (Selected Operational)	**Validation**	**0.40**	**77.20%**	**37.84%**	**71.79%**	**0.4956**	**28 / 39 (71.8%)**
Random Forest (Selected Operational)	**Held-out Test**	**0.40**	**78.00%**	**26.05%**	**58.49%**	**0.3605**	**31 / 53 (58.5%)**

6. Decision Threshold Policy & Asymmetric Cost Trade-Offs

Standard machine learning defaults to an arbitrary 0.50 probability cutoff. However, in port operations, **false negatives (unpredicted bottlenecks)** incur severe costs: idle containerships burning bunker fuel, intermodal rail/truck delays, and terminal demurrage fees reaching $20,000–$50,000 per day. Conversely, **false positives (false alarms)** merely trigger modest proactive review by dispatchers.

Validation threshold sweeping across 0.20 to 0.60 revealed distinct trade-offs: While threshold 0.55 maximizes validation F1 (0.5676), it misses nearly half of delayed cases (46.2% false negative rate). In contrast, **threshold 0.40** was selected as the operational baseline because it achieves **71.79% recall on validation** while preserving acceptable precision (37.84%). On the held-out test set, threshold 0.40 successfully detected **31 of 53 delays** (58.49% recall), compared to only 23 delays caught by threshold 0.50 and 14 delays caught by threshold 0.55.

7. Model Interpretability & Situational Operational Context

To build user trust without hallucinating precision, the system employs a two-tier interpretability framework:

- **Global Feature Weights:** Mean Decrease in Impunity (MDI) across all 100 trees identifies the primary factors governing model sensitivity: queue_length (0.2423), historical_avg_delay_hours (0.2064), and arrival_load_ratio (0.1603) explain over 60% of total variance, followed by vessel_arrivals_6h (0.0792) and effective_capacity_6h (0.0773).
- **Observed vs. Baseline Context:** For any given prediction, the system contrasts the user's observation against empirical training benchmarks (e.g., Queue: 8 vs. 1.04 baseline; Load Ratio: 1.33 vs. 0.74 baseline). This explains *operational pressure* without fabricating local attribution scores (such as ad-hoc SHAP approximations).
- **Governance Disclaimer:** The interface enforces an explicit non-causal notice: factors reflect historical statistical associations, not proven physical causation.

8. Interactive Prediction Interface (app.py)

An operator-facing Streamlit application provides accessible single-observation forecasting with enterprise safeguards:

- **Zero Retraining:** The app caches the pre-trained pipeline via @st.cache_resource, ensuring instant inference (<50ms) with zero model modification.
- **Automated Safe Math:** Automatically computes arrival_load_ratio = arrivals / capacity. If capacity is set to 0, it renders a validation warning and safely disables prediction, preventing division-by-zero crashes.
- **Synchronized Temporal Framing:** Weekday is automatically derived from the selected observation date (e.g. obs_date.strftime('%A')), and exact datetime arithmetic advances midnight boundaries into the next day.
- **Preset Operational Scenarios:** Features pre-loaded scenarios (*Normal Operations*, *Elevated Delay Risk*, *Borderline*) for immediate testing.

9. System Limitations & Future Technical Horizons

To maintain high integrity, the following limitations are formally documented:

- **Uncalibrated Probabilities:** Random Forest predicted probabilities represent ensemble tree voting fractions. While monotonic with empirical risk, they have not undergone isotonic or Platt probability calibration.
- **Synthetic Domain Gap:** The simulator abstracts away real-world complexities such as live AIS ship telemetry, tugboat availability, multi-crane outages, pilotage bottlenecks, and labor contract disputes.
- **Single-Step Horizon:** The model forecasts the immediate next 6-hour window. Multi-interval compounding delays over 24–48 hours will require autoregressive sequential architectures (e.g., Temporal Fusion Transformers).

Conclusion & Internship Readiness: The forecasting pipeline is reproducible, fully tested (17/17 automated tests passing), defensible in design, and backed by a responsive Streamlit dashboard. All code, data pipelines, and documentation adhere strictly to industry best practices in temporal machine learning governance.